lab1-logistyczna.R

Jamistrz

2024-06-22

```
logis <- read.csv("heart.csv")
View(logis)
library(lmtest)
```

```
## Ładowanie wymaganego pakietu: zoo
```

```
##
## Dołączanie pakietu: 'zoo'
```

```
## Następujące obiekty zostały zakryte z 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(doBy)
library(caret)
```

```
## Ładowanie wymaganego pakietu: ggplot2
```

```
## Ładowanie wymaganego pakietu: lattice
```

```
library(car)
```

```
## Ładowanie wymaganego pakietu: carData
```

```
library(verification)
```

```
## Ładowanie wymaganego pakietu: fields
```

```
## Ładowanie wymaganego pakietu: spam
```

```
## Spam version 2.10-0 (2023-10-23) is loaded.
## Type 'help( Spam)' or 'demo( spam)' for a short introduction
## and overview of this package.
## Help for individual functions is also obtained by adding the
## suffix '.spam' to the function name, e.g. 'help( chol.spam)'.
```

```
##
## Dołączanie pakietu: 'spam'
```

```
## Następujące obiekty zostały zakryte z 'package:base':
##
##     backsolve, forwardsolve
```

```
## Ładowanie wymaganego pakietu: viridisLite
```

```
##
## Try help(fields) to get started.
```

```
## Ładowanie wymaganego pakietu: boot
```

```
##
## Dołączanie pakietu: 'boot'
```

```
## Następujący obiekt został zakryty z 'package:car':
##
##     logit
```

```
## Następujący obiekt został zakryty z 'package:lattice':
##
##     melanoma
```

```
## Ładowanie wymaganego pakietu: CircStats
```

```
## Ładowanie wymaganego pakietu: MASS
```

```
## Ładowanie wymaganego pakietu: dtw
```

```
## Ładowanie wymaganego pakietu: proxy
```

```
##
## Dołączanie pakietu: 'proxy'
```

```
## Następujący obiekt został zakryty z 'package:spam':
##
##     as.matrix
```

```
## Następujące obiekty zostały zakryte z 'package:stats':
##
##     as.dist, dist
```

```
## Następujący obiekt został zakryty z 'package:base':
##
##     as.matrix
```

```
## Loaded dtw v1.23-1. See ?dtw for help, citation("dtw") for use in publication.
```

```
## Registered S3 method overwritten by 'verification':
##   method    from
##   lines.roc pROC
```

```
library(gmodels)
library(OptimalCutpoints)
library(ROCR)
library(randomForest)
```

```
## randomForest 4.7-1.1
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
## Dołączanie pakietu: 'randomForest'
```

```
## Następujący obiekt został zakryty z 'package:ggplot2':
##
##     margin
```

```
library(plotROC)
library(vcd)
```

```
## Ładowanie wymaganego pakietu: grid
```

```r
library(coin)
```

```
## Ładowanie wymaganego pakietu: survival
```

```
##
## Dołączanie pakietu: 'survival'
```

```
## Następujący obiekt został zakryty z 'package:verification':
##
##     brier
```

```
## Następujący obiekt został zakryty z 'package:boot':
##
##     aml
```

```
## Następujący obiekt został zakryty z 'package:caret':
##
##     cluster
```

```r
l<-ncol(logis)
l
```

```
## [1] 14
```

```r
X<-na.omit(data.frame(logis[1:14]))
factor(X$target)
```

```
##   [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [112] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [149] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [186] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [223] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [260] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
## [297] 0 0 0 0 0 0 0
## Levels: 0 1
```

```r
k<-ncol(X)
k
```

```
## [1] 14
```

```r
model0<-glm(X$target~.+0,data=X,family = "binomial"(link="logit"))
summary(model0)
```

```
##
## Call:
## glm(formula = X$target ~ . + 0, family = binomial(link = "logit"),
##     data = X)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## age       0.012771   0.019057   0.670 0.502772
## sex      -1.638052   0.451880  -3.625 0.000289 ***
## cp        0.849045   0.184063   4.613 3.97e-06 ***
## trestbps -0.015332   0.009815  -1.562 0.118265
## chol     -0.003615   0.003766  -0.960 0.337148
## fbs      -0.011519   0.525637  -0.022 0.982517
## restecg   0.543176   0.341823   1.589 0.112048
## thalach   0.031899   0.008441   3.779 0.000158 ***
## exang    -0.892011   0.402685  -2.215 0.026749 *
## oldpeak  -0.498808   0.209486  -2.381 0.017261 *
## slope     0.609170   0.345914   1.761 0.078231 .
## ca       -0.772491   0.189333  -4.080 4.50e-05 ***
## thal     -0.843846   0.287352  -2.937 0.003318 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 420.05  on 303  degrees of freedom
## Residual deviance: 213.27  on 290  degrees of freedom
## AIC: 239.27
##
## Number of Fisher Scoring iterations: 6
```

```r
# regresja krokowa
NAJ<-stepAIC(model0, direction = "both", steps = 1000, k = 2)
```

```
## Start:  AIC=239.27
## X$target ~ age + sex + cp + trestbps + chol + fbs + restecg +
##     thalach + exang + oldpeak + slope + ca + thal + 0
##
##            Df Deviance    AIC
## - fbs       1   213.27 237.27
## - age       1   213.72 237.72
## - chol      1   214.18 238.18
## <none>          213.27 239.27
## - trestbps  1   215.75 239.75
## - restecg   1   215.82 239.82
## - slope     1   216.32 240.32
## - exang     1   218.12 242.12
## - oldpeak   1   219.25 243.25
## - thal      1   222.15 246.15
## - sex       1   228.23 252.23
## - thalach   1   229.71 253.71
## - ca        1   231.00 255.00
## - cp        1   237.59 261.59
##
## Step:  AIC=237.27
## X$target ~ age + sex + cp + trestbps + chol + restecg + thalach +
##     exang + oldpeak + slope + ca + thal - 1
##
##            Df Deviance    AIC
## - age       1   213.73 235.73
## - chol      1   214.18 236.18
## <none>          213.27 237.27
## - trestbps  1   215.78 237.78
## - restecg   1   215.82 237.82
## - slope     1   216.37 238.37
## + fbs       1   213.27 239.27
## - exang     1   218.16 240.16
## - oldpeak   1   219.30 241.30
## - thal      1   222.47 244.47
## - sex       1   228.38 250.38
## - thalach   1   229.71 251.71
## - ca        1   231.31 253.31
## - cp        1   238.10 260.10
##
## Step:  AIC=235.73
## X$target ~ sex + cp + trestbps + chol + restecg + thalach + exang +
##     oldpeak + slope + ca + thal - 1
##
##            Df Deviance    AIC
## - chol      1   214.32 234.32
## <none>          213.73 235.73
## - trestbps  1   215.81 235.81
## - restecg   1   216.31 236.31
## - slope     1   216.94 236.94
## + age       1   213.27 237.27
## + fbs       1   213.72 237.72
```

```
## - exang     1   218.47 238.47
## - oldpeak   1   219.57 239.57
## - thal      1   222.53 242.53
## - sex       1   228.97 248.97
## - thalach   1   229.81 249.81
## - ca        1   231.37 251.37
## - cp        1   238.63 258.63
## 
## Step:  AIC=234.32
## X$target ~ sex + cp + trestbps + restecg + thalach + exang + 
##     oldpeak + slope + ca + thal - 1
## 
##            Df Deviance    AIC
## <none>          214.32 234.32
## - slope     1   217.35 235.35
## - restecg   1   217.38 235.38
## - trestbps  1   217.39 235.39
## + chol      1   213.73 235.73
## + age       1   214.18 236.18
## + fbs       1   214.32 236.32
## - exang     1   219.38 237.38
## - oldpeak   1   220.91 238.91
## - thal      1   224.51 242.51
## - sex       1   229.07 247.07
## - thalach   1   230.30 248.30
## - ca        1   232.15 250.15
## - cp        1   239.90 257.90
```

```
NAJ
```

```
## 
## Call:  glm(formula = X$target ~ sex + cp + trestbps + restecg + thalach + 
##     exang + oldpeak + slope + ca + thal - 1, family = binomial(link = "logit"), 
##     data = X)
## 
## Coefficients:
##      sex        cp  trestbps   restecg   thalach     exang   oldpeak     slope  
## -1.54470   0.85741  -0.01394   0.58311   0.02907  -0.89737  -0.51230   0.60240  
##       ca      thal  
## -0.74795  -0.85674  
## 
## Degrees of Freedom: 303 Total (i.e. Null);  293 Residual
## Null Deviance:       420 
## Residual Deviance: 214.3     AIC: 234.3
```

```
summary(NAJ)
```

```
## 
## Call:
## glm(formula = X$target ~ sex + cp + trestbps + restecg + thalach + 
##     exang + oldpeak + slope + ca + thal - 1, family = binomial(link = "logit"), 
##     data = X)
## 
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)    
## sex      -1.544700   0.428317  -3.606 0.000310 ***
## cp        0.857412   0.180838   4.741 2.12e-06 ***
## trestbps -0.013938   0.008056  -1.730 0.083613 .  
## restecg   0.583114   0.335211   1.740 0.081939 .  
## thalach   0.029072   0.007707   3.772 0.000162 ***
## exang    -0.897368   0.396991  -2.260 0.023795 *  
## oldpeak  -0.512303   0.205526  -2.493 0.012680 *  
## slope     0.602397   0.342627   1.758 0.078718 .  
## ca       -0.747949   0.184840  -4.046 5.20e-05 ***
## thal     -0.856743   0.272489  -3.144 0.001666 ** 
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## (Dispersion parameter for binomial family taken to be 1)
## 
##     Null deviance: 420.05  on 303  degrees of freedom
## Residual deviance: 214.32  on 293  degrees of freedom
## AIC: 234.32
## 
## Number of Fisher Scoring iterations: 6
```

```
print("najlepsze z wyrazem wolnym")
```

```
## [1] "najlepsze z wyrazem wolnym"
```

```
model2<-glm(X$target~.,data=X,family = "binomial")
NAJ1<-stepAIC(model2, direction = "both", steps = 1000, k = 2)
```

```
## Start:  AIC=239.44
## X$target ~ age + sex + cp + trestbps + chol + fbs + restecg + 
##     thalach + exang + oldpeak + slope + ca + thal
## 
##            Df Deviance    AIC
## - fbs       1   211.44 237.44
## - age       1   211.48 237.48
## - chol      1   212.91 238.91
## - restecg   1   213.24 239.24
## <none>          211.44 239.44
## - slope     1   214.13 240.13
## - trestbps  1   215.08 241.08
## - thalach   1   216.64 242.64
## - exang     1   217.09 243.09
## - oldpeak   1   218.24 244.24
## - thal      1   221.36 247.36
## - sex       1   227.77 253.77
## - ca        1   228.84 254.84
## - cp        1   236.18 262.18
## 
## Step:  AIC=237.44
## X$target ~ age + sex + cp + trestbps + chol + restecg + thalach + 
##     exang + oldpeak + slope + ca + thal
## 
##            Df Deviance    AIC
## - age       1   211.48 235.48
## - chol      1   212.91 236.91
## - restecg   1   213.24 237.24
## <none>          211.44 237.44
## - slope     1   214.14 238.14
## - trestbps  1   215.10 239.10
## + fbs       1   211.44 239.44
## - thalach   1   216.67 240.67
## - exang     1   217.11 241.11
## - oldpeak   1   218.33 242.33
## - thal      1   221.80 245.80
## - sex       1   227.90 251.90
## - ca        1   229.09 253.09
## - cp        1   236.84 260.84
## 
## Step:  AIC=235.48
## X$target ~ sex + cp + trestbps + chol + restecg + thalach + exang + 
##     oldpeak + slope + ca + thal
## 
##            Df Deviance    AIC
## - chol      1   213.08 235.08
## - restecg   1   213.34 235.34
## <none>          211.48 235.48
## - slope     1   214.19 236.19
## + age       1   211.44 237.44
## + fbs       1   211.48 237.48
## - trestbps  1   215.57 237.57
## - exang     1   217.12 239.12
## - thalach   1   218.29 240.29
```

```
## - oldpeak   1   218.34 240.34
## - thal      1   221.86 243.86
## - sex       1   227.96 249.96
## - ca        1   230.16 252.16
## - cp        1   236.84 258.84
##
## Step:  AIC=235.08
## X$target ~ sex + cp + trestbps + restecg + thalach + exang +
##     oldpeak + slope + ca + thal
##
##            Df Deviance    AIC
## <none>          213.08 235.08
## + chol      1   211.48 235.48
## - slope     1   215.68 235.68
## - restecg   1   215.77 235.77
## + age       1   212.91 236.91
## + fbs       1   213.08 237.08
## - trestbps  1   217.38 237.38
## - exang     1   218.81 238.81
## - thalach   1   219.37 239.37
## - oldpeak   1   220.55 240.55
## - thal      1   224.46 244.46
## - sex       1   227.96 247.96
## - ca        1   231.65 251.65
## - cp        1   239.20 259.20
```

```
NAJ1
```

```
##
## Call:  glm(formula = X$target ~ sex + cp + trestbps + restecg + thalach +
##     exang + oldpeak + slope + ca + thal, family = "binomial",
##     data = X)
##
## Coefficients:
## (Intercept)          sex           cp     trestbps      restecg      thalach
##     2.15791     -1.56359      0.87048     -0.02041      0.55340      0.02281
##       exang      oldpeak        slope           ca         thal
##    -0.97066     -0.55861      0.56462     -0.76304     -0.94044
##
## Degrees of Freedom: 302 Total (i.e. Null);  292 Residual
## Null Deviance:       417.6
## Residual Deviance: 213.1     AIC: 235.1
```

```
summary(NAJ1)
```

```
##
## Call:
## glm(formula = X$target ~ sex + cp + trestbps + restecg + thalach +
##     exang + oldpeak + slope + ca + thal, family = "binomial",
##     data = X)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.157907   1.945343   1.109 0.267315
## sex         -1.563590   0.432720  -3.613 0.000302 ***
## cp           0.870475   0.182131   4.779 1.76e-06 ***
## trestbps    -0.020406   0.009988  -2.043 0.041046 *
## restecg      0.553402   0.339071   1.632 0.102656
## thalach      0.022809   0.009395   2.428 0.015185 *
## exang       -0.970656   0.403365  -2.406 0.016111 *
## oldpeak     -0.558608   0.211713  -2.639 0.008327 **
## slope        0.564616   0.346525   1.629 0.103236
## ca          -0.763038   0.185060  -4.123 3.74e-05 ***
## thal        -0.940441   0.284072  -3.311 0.000931 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 417.64  on 302  degrees of freedom
## Residual deviance: 213.08  on 292  degrees of freedom
## AIC: 235.08
##
## Number of Fisher Scoring iterations: 6
```

```
#print("kryterium AKaike lepszy model to model z najmniejsze AIC")
AIC(NAJ,NAJ1)
```

```
##      df      AIC
## NAJ  10 234.3206
## NAJ1 11 235.0768
```

```
print("analizy dla modelu NAJ")
```

```
## [1] "analizy dla modelu NAJ"
```

```
S<-summary(NAJ)
S
```

```
##
## Call:
## glm(formula = X$target ~ sex + cp + trestbps + restecg + thalach +
##     exang + oldpeak + slope + ca + thal - 1, family = binomial(link = "logit"),
##     data = X)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## sex      -1.544700   0.428317  -3.606 0.000310 ***
## cp        0.857412   0.180838   4.741 2.12e-06 ***
## trestbps -0.013938   0.008056  -1.730 0.083613 .
## restecg   0.583114   0.335211   1.740 0.081939 .
## thalach   0.029072   0.007707   3.772 0.000162 ***
## exang    -0.897368   0.396991  -2.260 0.023795 *
## oldpeak  -0.512303   0.205526  -2.493 0.012680 *
## slope     0.602397   0.342627   1.758 0.078718 .
## ca       -0.747949   0.184840  -4.046 5.20e-05 ***
## thal     -0.856743   0.272489  -3.144 0.001666 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 420.05  on 303  degrees of freedom
## Residual deviance: 214.32  on 293  degrees of freedom
## AIC: 234.32
##
## Number of Fisher Scoring iterations: 6
```

```
W1<-data.frame(round(S$coefficients[,4],5))
pvalue<-W1$round.S.coefficients...4...5.
"przedzia?y ufno?ci dla wsp??czynnik?W"
```

```
## [1] "przedzia?y ufno?ci dla wsp??czynnik?W"
```

```
CON<-Confint(NAJ)
W2<-data.frame(CON,pvalue)
colnames(W2)<-c("coeficient","95% CI lower","95% CI upper","p-value")
W2
```

```
##          coeficient 95% CI lower 95% CI upper p-value
## sex      -1.54469998  -2.42450747 -0.735755118 0.00031
## cp        0.85741175   0.51410327  1.226406548 0.00000
## trestbps -0.01393815  -0.03010583  0.001630877 0.08361
## restecg   0.58311354  -0.06960147  1.250260883 0.08194
## thalach   0.02907192   0.01445247  0.044790834 0.00016
## exang    -0.89736766  -1.68091820 -0.116362498 0.02380
## oldpeak  -0.51230311  -0.92880995 -0.118994958 0.01268
## slope     0.60239673  -0.07665157  1.275582710 0.07872
## ca       -0.74794899  -1.12475548 -0.394867941 0.00005
## thal     -0.85674317  -1.40439863 -0.329934740 0.00167
```

```
"Relativ variable important"
```

```
## [1] "Relativ variable important"
```

```
RIV<-varImp(NAJ, scale = TRUE)
RIV
```

```
##          Overall
## cp       4.741313
## trestbps 1.730096
## restecg  1.739543
## thalach  3.772214
## exang    2.260421
## oldpeak  2.492638
## slope    1.758172
## ca       4.046475
## thal     3.144136
```

```
"badanie wsp??iniowo?ci w modelu z wyrazem wolnym
 vif<10  brak wsp?liniowo?ci zmiennych"
```

```
## [1] "badanie wsp??iniowo?ci w modelu z wyrazem wolnym\n vif<10  brak wsp?liniowo?ci zmiennych"
```

```
vif(NAJ1)
```

```
##      sex       cp trestbps  restecg  thalach    exang  oldpeak    slope
## 1.165703 1.245034 1.065543 1.039551 1.192727 1.123495 1.402143 1.487628
##       ca     thal
## 1.073931 1.038523
```

```
#testy LR dopasowania modelu
LR<-lrtest(NAJ)
LR
```

```
## Likelihood ratio test
##
## Model 1: X$target ~ sex + cp + trestbps + restecg + thalach + exang +
##     oldpeak + slope + ca + thal - 1
## Model 2: X$target ~ 1
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1  10 -107.16
## 2   1 -208.82 -9 203.32  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ifelse(LR$`Pr(>Chisq)`<0.05,"model dobrze dopasowany","model ?le dopasowany")
```

```
## [1] NA                        "model dobrze dopasowany"
```

```
LR$`Pr(>Chisq)`[[2]]
```

```
## [1] 6.677398e-39
```

```
WALD<-waldtest(NAJ)
WALD
```

```
## Wald test
##
## Model 1: X$target ~ sex + cp + trestbps + restecg + thalach + exang +
##     oldpeak + slope + ca + thal - 1
## Model 2: X$target ~ 1 - 1
##   Res.Df  Df      F    Pr(>F)
## 1    293
## 2    303 -10 8.1141 1.404e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

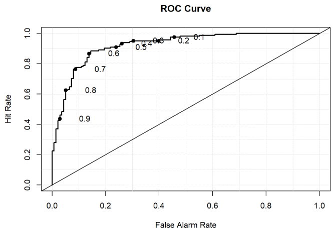
```
WALD$`Pr(>F)`[[2]]
```

```
## [1] 1.403888e-11
```

```
ifelse(WALD$`Pr(>F)`<0.05,"istotny model","tylko istotna stała")
```

```
## [1] NA              "istotny model"
```

```
roc.plot(NAJ$y,NAJ$fitted.values)
```



```
AUC<-roc.area(NAJ1$y,NAJ1$fitted.values)
AUC
```

```
## $A
## [1] 0.9232323
##
## $n.total
## [1] 303
##
## $n.events
## [1] 165
##
## $n.noevents
## [1] 138
##
## $p.value
## [1] 3.448073e-37
```

```
#od ratio
ILORAZYSZANS<-exp(NAJ$coefficients)
ILORAZYSZANS
```

```
##       sex        cp  trestbps   restecg   thalach     exang   oldpeak     slope
## 0.2133759 2.3570522 0.9861585 1.7916080 1.0294986 0.4076413 0.5991142 1.8264912
##        ca      thal
## 0.4733364 0.4245425
```

```
print("predykcja z modelu dla ewentualnie innych danych")
```

```
## [1] "predykcja z modelu dla ewentualnie innych danych"
```

```
pred<-predict.glm(NAJ,X,type = "response")
pred
```

```
##           1           2           3           4           5           6
## 0.791212784 0.704991088 0.943742046 0.922693132 0.873891797 0.717166399
##           7           8           9          10          11          12
## 0.828777946 0.868442073 0.809063856 0.916265182 0.649108606 0.980465035
##          13          14          15          16          17          18
## 0.903705259 0.679198862 0.984785395 0.964013034 0.994041953 0.791394352
##          19          20          21          22          23          24
## 0.655169725 0.935059056 0.404525992 0.926502793 0.852353648 0.531111579
##          25          26          27          28          29          30
## 0.864605628 0.846908950 0.869713533 0.878661128 0.934703426 0.610364809
##          31          32          33          34          35          36
## 0.973518242 0.466437345 0.921114039 0.532255114 0.512003374 0.742506125
##          37          38          39          40          41          42
## 0.992231046 0.666222019 0.971297057 0.950548375 0.866064229 0.820560250
##          43          44          45          46          47          48
## 0.096477617 0.736782942 0.952627374 0.931659462 0.949933289 0.906631800
##          49          50          51          52          53          54
## 0.988780340 0.901973719 0.969814284 0.464204154 0.124766433 0.988589347
##          55          56          57          58          59          60
## 0.986937366 0.722118073 0.839367594 0.848389527 0.980784867 0.829446611
##          61          62          63          64          65          66
## 0.937281950 0.826394785 0.990549125 0.831643786 0.924628066 0.938475497
##          67          68          69          70          71          72
## 0.709608992 0.937880755 0.934427707 0.956261939 0.645281900 0.712872076
##          73          74          75          76          77          78
## 0.946067848 0.623698359 0.984943871 0.861526496 0.713649752 0.786879736
##          79          80          81          82          83          84
## 0.950371266 0.502777804 0.979912101 0.876766769 0.986226115 0.889363445
##          85          86          87          88          89          90
## 0.669652011 0.878246188 0.705308717 0.839949963 0.968547818 0.629382032
##          91          92          93          94          95          96
## 0.891668824 0.455085592 0.560314411 0.815477414 0.941601282 0.071775006
##          97          98          99         100         101         102
## 0.708290780 0.135421782 0.805481190 0.653637391 0.848779148 0.123474673
##         103         104         105         106         107         108
## 0.936342686 0.850755617 0.960275266 0.818463505 0.667207947 0.594969886
##         109         110         111         112         113         114
## 0.967872009 0.929599016 0.758644391 0.813984485 0.939645367 0.694312821
##         115         116         117         118         119         120
## 0.889074347 0.993687511 0.751485538 0.754618800 0.988667654 0.619401811
##         121         122         123         124         125         126
## 0.211874128 0.788220473 0.976979052 0.989603357 0.996607980 0.989124713
##         127         128         129         130         131         132
## 0.745490130 0.980603333 0.972617049 0.609703485 0.972075099 0.959742735
##         133         134         135         136         137         138
## 0.918654579 0.909037570 0.980436813 0.952627600 0.948219408 0.748382589
##         139         140         141         142         143         144
## 0.320409689 0.032023802 0.977913666 0.725456365 0.989478027 0.745660544
##         145         146         147         148         149         150
## 0.932606157 0.687176519 0.951303317 0.993732927 0.970261435 0.942316651
##         151         152         153         154         155         156
## 0.355917966 0.673763430 0.708680217 0.903762452 0.975453057 0.761533292
##         157         158         159         160         161         162
## 0.974330788 0.939641260 0.056007279 0.705585506 0.805795091 0.964537461
##         163         164         165         166         167         168
## 0.952828458 0.588731967 0.588731967 0.003485002 0.005708429 0.086787249
##         169         170         171         172         173         174
## 0.074737657 0.017197738 0.569382403 0.540952793 0.564244395 0.242141594
##         175         176         177         178         179         180
## 0.006001109 0.025246326 0.081902908 0.947178371 0.020340209 0.070018033
##         181         182         183         184         185         186
## 0.039473885 0.029297824 0.930377477 0.356492192 0.038640870 0.521384267
##         187         188         189         190         191         192
## 0.089874652 0.007810792 0.626552595 0.537437555 0.358795150 0.003152268
##         193         194         195         196         197         198
## 0.058302258 0.005308802 0.519223348 0.006346126 0.495176673 0.177800135
##         199         200         201         202         203         204
## 0.006436658 0.515054824 0.692363393 0.014267145 0.043894235 0.159381017
##         205         206         207         208         209         210
## 0.002460244 0.254293486 0.040017552 0.089305064 0.107682984 0.228988951
##         211         212         213         214         215         216
## 0.456640546 0.017609746 0.246231022 0.239752868 0.077860187 0.092107940
##         217         218         219         220         221         222
## 0.493852386 0.007047834 0.020145033 0.059828902 0.020341540 0.002417891
##         223         224         225         226         227         228
## 0.830097987 0.002641477 0.020246501 0.015547078 0.057407206 0.073121493
##         229         230         231         232         233         234
## 0.770332875 0.275288082 0.959228891 0.006939516 0.027057325 0.007302252
##         235         236         237         238         239         240
## 0.008390043 0.275683737 0.235591268 0.067087799 0.097332174 0.263376776
##         241         242         243         244         245         246
## 0.034903522 0.576080244 0.075641923 0.008579446 0.035315342 0.338255946
##         247         248         249         250         251         252
## 0.055127657 0.096110432 0.547909185 0.058800597 0.001322545 0.005849818
##         253         254         255         256         257         258
## 0.069977170 0.036616490 0.872442380 0.012624733 0.004288839 0.038604952
##         259         260         261         262         263         264
## 0.560646242 0.602815008 0.121915806 0.705204571 0.004967538 0.392801153
##         265         266         267         268         269         270
## 0.063296586 0.186408323 0.155576951 0.274104132 0.006583806 0.009475418
##         271         272         273         274         275         276
## 0.308700864 0.504847238 0.100340147 0.503800659 0.051362462 0.232649690
##         277         278         279         280         281         282
## 0.024497236 0.678635588 0.798374290 0.011717651 0.214662848 0.727559588
##         283         284         285         286         287         288
## 0.682070914 0.693470861 0.030229402 0.008946191 0.884745220 0.663155881
##         289         290         291         292         293         294
## 0.029668353 0.185992676 0.387641151 0.037252329 0.099119546 0.508658524
##         295         296         297         298         299         300
## 0.162959962 0.005954720 0.689997348 0.029984503 0.318637083 0.791228983
##         301         302         303
## 0.016706652 0.025131312 0.904212129
```

```
DF<- data.frame(fitted(NAJ),NAJ$y)
DF
```

```
##     fitted.NAJ. NAJ.y
## 1   0.791212784     1
## 2   0.704991088     1
## 3   0.943742046     1
## 4   0.922693132     1
## 5   0.873891797     1
## 6   0.717166399     1
## 7   0.828777946     1
## 8   0.868442073     1
## 9   0.809063856     1
## 10  0.916265182     1
## 11  0.649108606     1
## 12  0.980465035     1
## 13  0.903705259     1
## 14  0.679198862     1
## 15  0.984785395     1
## 16  0.964013034     1
## 17  0.994041953     1
## 18  0.791394352     1
## 19  0.655169725     1
## 20  0.935059056     1
## 21  0.404525992     1
## 22  0.926502793     1
## 23  0.852353648     1
## 24  0.531111579     1
## 25  0.864605628     1
## 26  0.846908950     1
## 27  0.869713533     1
## 28  0.878661128     1
## 29  0.934703426     1
## 30  0.610364809     1
## 31  0.973518242     1
## 32  0.466437345     1
## 33  0.921114039     1
## 34  0.532255114     1
## 35  0.512003374     1
## 36  0.742506125     1
## 37  0.992231046     1
## 38  0.666222019     1
## 39  0.971297057     1
## 40  0.950548375     1
## 41  0.866064229     1
## 42  0.820560250     1
## 43  0.096477617     1
## 44  0.736782942     1
## 45  0.952627374     1
## 46  0.931659462     1
## 47  0.949933289     1
## 48  0.906631800     1
## 49  0.988780340     1
## 50  0.901973719     1
## 51  0.969814284     1
## 52  0.464204154     1
## 53  0.124766433     1
## 54  0.988589347     1
## 55  0.986937366     1
## 56  0.722118073     1
## 57  0.839367594     1
## 58  0.848389527     1
## 59  0.980784867     1
## 60  0.829446611     1
## 61  0.937281950     1
## 62  0.826394785     1
## 63  0.990549125     1
## 64  0.831643786     1
## 65  0.924628066     1
## 66  0.938475497     1
## 67  0.709608992     1
## 68  0.937880755     1
## 69  0.934427707     1
## 70  0.956261939     1
## 71  0.645281900     1
## 72  0.712872076     1
## 73  0.946067848     1
## 74  0.623698359     1
## 75  0.984943871     1
## 76  0.861526496     1
## 77  0.713649752     1
## 78  0.786879736     1
## 79  0.950371266     1
## 80  0.502777804     1
## 81  0.979912101     1
## 82  0.876766769     1
## 83  0.986226115     1
## 84  0.889363445     1
## 85  0.669652011     1
## 86  0.878246188     1
## 87  0.705308717     1
## 88  0.839949963     1
## 89  0.968547818     1
## 90  0.629382032     1
## 91  0.891668824     1
## 92  0.455085592     1
## 93  0.560314411     1
## 94  0.815477414     1
## 95  0.941601282     1
## 96  0.071775006     1
## 97  0.708290780     1
## 98  0.135421782     1
## 99  0.805481190     1
## 100 0.653637391     1
## 101 0.848779148     1
## 102 0.123474673     1
## 103 0.936342686     1
## 104 0.850755617     1
## 105 0.960275266     1
## 106 0.818463505     1
## 107 0.667207947     1
## 108 0.594969886     1
## 109 0.967872009     1
## 110 0.929599016     1
## 111 0.758644391     1
## 112 0.813984485     1
## 113 0.939645367     1
## 114 0.694312821     1
## 115 0.889074347     1
## 116 0.993687511     1
## 117 0.751485538     1
## 118 0.754618800     1
## 119 0.988667654     1
## 120 0.619401811     1
## 121 0.211874128     1
## 122 0.788220473     1
## 123 0.976979052     1
## 124 0.989603357     1
## 125 0.996607980     1
## 126 0.989124713     1
## 127 0.745490130     1
## 128 0.980603333     1
## 129 0.972617049     1
## 130 0.609703485     1
## 131 0.972075099     1
## 132 0.959742735     1
## 133 0.918654579     1
## 134 0.909037570     1
## 135 0.980436813     1
## 136 0.952627600     1
## 137 0.948219408     1
## 138 0.748382589     1
## 139 0.320409689     1
## 140 0.032023802     1
## 141 0.977913666     1
```

```
## 142 0.725456365     1
## 143 0.989478027     1
## 144 0.745660544     1
## 145 0.932606157     1
## 146 0.687176519     1
## 147 0.951303317     1
## 148 0.993732927     1
## 149 0.970261435     1
## 150 0.942316651     1
## 151 0.355917966     1
## 152 0.673763430     1
## 153 0.708680217     1
## 154 0.903762452     1
## 155 0.975453057     1
## 156 0.761533292     1
## 157 0.974330788     1
## 158 0.939641260     1
## 159 0.056007279     1
## 160 0.705585506     1
## 161 0.805795091     1
## 162 0.964537461     1
## 163 0.952828458     1
## 164 0.588731967     1
## 165 0.588731967     1
## 166 0.003485002     0
## 167 0.005708429     0
## 168 0.086787249     0
## 169 0.074737657     0
## 170 0.017197738     0
## 171 0.569382403     0
## 172 0.540952793     0
## 173 0.564244395     0
## 174 0.242141594     0
## 175 0.006001109     0
## 176 0.025246326     0
## 177 0.081902908     0
## 178 0.947178371     0
## 179 0.020340209     0
## 180 0.070018033     0
## 181 0.039473885     0
## 182 0.029297824     0
## 183 0.930377477     0
## 184 0.356492192     0
## 185 0.038640870     0
## 186 0.521384267     0
## 187 0.089874652     0
## 188 0.007810792     0
## 189 0.626552595     0
## 190 0.537437555     0
## 191 0.358795150     0
## 192 0.003152268     0
## 193 0.058302258     0
## 194 0.005308802     0
## 195 0.519223348     0
## 196 0.006346126     0
## 197 0.495176673     0
## 198 0.177800135     0
## 199 0.006436658     0
## 200 0.515054824     0
## 201 0.692363393     0
## 202 0.014267145     0
## 203 0.043894235     0
## 204 0.159381017     0
## 205 0.002460244     0
## 206 0.254293486     0
## 207 0.040017552     0
## 208 0.089305064     0
## 209 0.107682984     0
## 210 0.228988951     0
## 211 0.456640546     0
## 212 0.017609746     0
## 213 0.246231022     0
## 214 0.239752868     0
## 215 0.077860187     0
## 216 0.092107940     0
## 217 0.493852386     0
## 218 0.007047834     0
## 219 0.020145033     0
## 220 0.059828902     0
## 221 0.020341540     0
## 222 0.002417891     0
## 223 0.830097987     0
## 224 0.002641477     0
## 225 0.020246501     0
## 226 0.015547078     0
## 227 0.057407206     0
## 228 0.073121493     0
## 229 0.770332875     0
## 230 0.275288082     0
## 231 0.959228891     0
## 232 0.006939516     0
## 233 0.027057325     0
## 234 0.007302252     0
## 235 0.008390043     0
## 236 0.275683737     0
## 237 0.235591268     0
## 238 0.067087799     0
## 239 0.097332174     0
## 240 0.263376776     0
## 241 0.034903522     0
## 242 0.576080244     0
## 243 0.075641923     0
## 244 0.008579446     0
## 245 0.035315342     0
## 246 0.338255946     0
## 247 0.055127657     0
## 248 0.096110432     0
## 249 0.547909185     0
## 250 0.058800597     0
## 251 0.001322545     0
## 252 0.005849818     0
## 253 0.069977170     0
## 254 0.036616490     0
## 255 0.872442380     0
## 256 0.012624733     0
## 257 0.004288839     0
## 258 0.038604952     0
## 259 0.560646242     0
## 260 0.602815008     0
## 261 0.121915806     0
## 262 0.705204571     0
## 263 0.004967538     0
## 264 0.392801153     0
## 265 0.063296586     0
## 266 0.186408323     0
## 267 0.155576951     0
## 268 0.274104132     0
## 269 0.006583806     0
## 270 0.009475418     0
## 271 0.308700864     0
## 272 0.504847238     0
## 273 0.100340147     0
## 274 0.503800659     0
## 275 0.051362462     0
## 276 0.232649690     0
## 277 0.024497236     0
## 278 0.678635588     0
## 279 0.798374290     0
## 280 0.011717651     0
## 281 0.214662848     0
## 282 0.727559588     0
## 283 0.682070914     0
## 284 0.693470861     0
```

```
## 285 0.030229402     0
## 286 0.008946191     0
## 287 0.884745220     0
## 288 0.663155881     0
## 289 0.029668353     0
## 290 0.185992676     0
## 291 0.387641151     0
## 292 0.037252329     0
## 293 0.099119546     0
## 294 0.508658524     0
## 295 0.162959962     0
## 296 0.005954720     0
## 297 0.689997348     0
## 298 0.029984503     0
## 299 0.318637083     0
## 300 0.791228983     0
## 301 0.016706652     0
## 302 0.025131312     0
## 303 0.904212129     0
```

```r
#podsumowanie
CrossTable(NAJ$y,fitted(NAJ)>0.05,prop.t = T,prop.chisq = T)
```

```
## 
## 
##    Cell Contents
## |-------------------------|
## |                       N |
## | Chi-square contribution |
## |           N / Row Total |
## |           N / Col Total |
## |         N / Table Total |
## |-------------------------|
## 
## 
## Total Observations in Table:  303 
## 
## 
##              | fitted(NAJ) > 0.05 
##        NAJ$y |     FALSE |      TRUE | Row Total | 
## -------------|-----------|-----------|-----------|
##            0 |        52 |        86 |       138 | 
##              |    32.158 |     6.818 |           | 
##              |     0.377 |     0.623 |     0.455 | 
##              |     0.981 |     0.344 |           | 
##              |     0.172 |     0.284 |           | 
## -------------|-----------|-----------|-----------|
##            1 |         1 |       164 |       165 | 
##              |    26.896 |     5.702 |           | 
##              |     0.006 |     0.994 |     0.545 | 
##              |     0.019 |     0.656 |           | 
##              |     0.003 |     0.541 |           | 
## -------------|-----------|-----------|-----------|
## Column Total |        53 |       250 |       303 | 
##              |     0.175 |     0.825 |           | 
## -------------|-----------|-----------|-----------|
## 
## 
```

```r
A1<-matrix(0,nrow(DF),1)
A2<-data.frame(A1)
colnames(A2)<-c("predict")
for(i in 1:nrow(DF)){
  if(DF[i,1]>=0.60){A2[i,1]=0}
  if(DF[i,1]>=0.60){A2[i,1]=1}
}
A2
```

```
##     predict
## 1         1
## 2         1
## 3         1
## 4         1
## 5         1
## 6         1
## 7         1
## 8         1
## 9         1
## 10        1
## 11        1
## 12        1
## 13        1
## 14        1
## 15        1
## 16        1
## 17        1
## 18        1
## 19        1
## 20        1
## 21        0
## 22        1
## 23        1
## 24        0
## 25        1
## 26        1
## 27        1
## 28        1
## 29        1
## 30        1
## 31        1
## 32        0
## 33        1
## 34        0
## 35        0
## 36        1
## 37        1
## 38        1
## 39        1
## 40        1
## 41        1
## 42        1
## 43        0
## 44        1
## 45        1
## 46        1
## 47        1
## 48        1
## 49        1
## 50        1
## 51        1
## 52        0
## 53        0
## 54        1
## 55        1
## 56        1
## 57        1
## 58        1
## 59        1
## 60        1
## 61        1
## 62        1
## 63        1
## 64        1
## 65        1
## 66        1
## 67        1
## 68        1
## 69        1
## 70        1
## 71        1
```

```
## 72        1
## 73        1
## 74        1
## 75        1
## 76        1
## 77        1
## 78        1
## 79        1
## 80        0
## 81        1
## 82        1
## 83        1
## 84        1
## 85        1
## 86        1
## 87        1
## 88        1
## 89        1
## 90        1
## 91        1
## 92        0
## 93        0
## 94        1
## 95        1
## 96        0
## 97        1
## 98        0
## 99        1
## 100       1
## 101       1
## 102       0
## 103       1
## 104       1
## 105       1
## 106       1
## 107       1
## 108       0
## 109       1
## 110       1
## 111       1
## 112       1
## 113       1
## 114       1
## 115       1
## 116       1
## 117       1
## 118       1
## 119       1
## 120       1
## 121       0
## 122       1
## 123       1
## 124       1
## 125       1
## 126       1
## 127       1
## 128       1
## 129       1
## 130       1
## 131       1
## 132       1
## 133       1
## 134       1
## 135       1
## 136       1
## 137       1
## 138       1
## 139       0
## 140       0
## 141       1
## 142       1
## 143       1
## 144       1
## 145       1
## 146       1
## 147       1
## 148       1
## 149       1
## 150       1
## 151       0
## 152       1
## 153       1
## 154       1
## 155       1
## 156       1
## 157       1
## 158       1
## 159       0
## 160       1
## 161       1
## 162       1
## 163       1
## 164       0
## 165       0
## 166       0
## 167       0
## 168       0
## 169       0
## 170       0
## 171       0
## 172       0
## 173       0
## 174       0
## 175       0
## 176       0
## 177       0
## 178       1
## 179       0
## 180       0
## 181       0
## 182       0
## 183       1
## 184       0
## 185       0
## 186       0
## 187       0
## 188       0
## 189       1
## 190       0
## 191       0
## 192       0
## 193       0
## 194       0
## 195       0
## 196       0
## 197       0
## 198       0
## 199       0
## 200       0
## 201       1
## 202       0
## 203       0
## 204       0
## 205       0
## 206       0
## 207       0
## 208       0
## 209       0
## 210       0
## 211       0
## 212       0
## 213       0
## 214       0
```

```
## 215       0
## 216       0
## 217       0
## 218       0
## 219       0
## 220       0
## 221       0
## 222       0
## 223       1
## 224       0
## 225       0
## 226       0
## 227       0
## 228       0
## 229       1
## 230       0
## 231       1
## 232       0
## 233       0
## 234       0
## 235       0
## 236       0
## 237       0
## 238       0
## 239       0
## 240       0
## 241       0
## 242       0
## 243       0
## 244       0
## 245       0
## 246       0
## 247       0
## 248       0
## 249       0
## 250       0
## 251       0
## 252       0
## 253       0
## 254       0
## 255       1
## 256       0
## 257       0
## 258       0
## 259       0
## 260       1
## 261       0
## 262       1
## 263       0
## 264       0
## 265       0
## 266       0
## 267       0
## 268       0
## 269       0
## 270       0
## 271       0
## 272       0
## 273       0
## 274       0
## 275       0
## 276       0
## 277       0
## 278       1
## 279       1
## 280       0
## 281       0
## 282       1
## 283       1
## 284       1
## 285       0
## 286       0
## 287       1
## 288       1
## 289       0
## 290       0
## 291       0
## 292       0
## 293       0
## 294       0
## 295       0
## 296       0
## 297       1
## 298       0
## 299       0
## 300       1
## 301       0
## 302       0
## 303       1
```

```
DF1<-data.frame(DF,A2)
DF1
```

```
##     fitted.NAJ. NAJ.y predict
## 1   0.791212784     1       1
## 2   0.704991088     1       1
## 3   0.943742046     1       1
## 4   0.922693132     1       1
## 5   0.873891797     1       1
## 6   0.717166399     1       1
## 7   0.828777946     1       1
## 8   0.868442073     1       1
## 9   0.809063856     1       1
## 10  0.916265182     1       1
## 11  0.649108606     1       1
## 12  0.980465035     1       1
## 13  0.903705259     1       1
## 14  0.679198862     1       1
## 15  0.984785395     1       1
## 16  0.964013034     1       1
## 17  0.994041953     1       1
## 18  0.791394352     1       1
## 19  0.655169725     1       1
## 20  0.935059056     1       1
## 21  0.404525992     1       0
## 22  0.926502793     1       1
## 23  0.852353648     1       1
## 24  0.531111579     1       0
## 25  0.864605628     1       1
## 26  0.846908950     1       1
## 27  0.869713533     1       1
## 28  0.878661128     1       1
## 29  0.934703426     1       1
## 30  0.610364809     1       1
## 31  0.973518242     1       1
## 32  0.466437345     1       0
## 33  0.921114039     1       1
## 34  0.532255114     1       0
## 35  0.512003374     1       0
## 36  0.742506125     1       1
## 37  0.992231046     1       1
## 38  0.666222019     1       1
## 39  0.971297057     1       1
## 40  0.950548375     1       1
## 41  0.866064229     1       1
## 42  0.820560250     1       1
## 43  0.096477617     1       0
## 44  0.736782942     1       1
## 45  0.952627374     1       1
## 46  0.931659462     1       1
## 47  0.949933289     1       1
```

```
## 48  0.906631800     1       1
## 49  0.988780340     1       1
## 50  0.901973719     1       1
## 51  0.969814284     1       1
## 52  0.464204154     1       0
## 53  0.124766433     1       0
## 54  0.988589347     1       1
## 55  0.986937366     1       1
## 56  0.722118073     1       1
## 57  0.839367594     1       1
## 58  0.848389527     1       1
## 59  0.980784867     1       1
## 60  0.829446611     1       1
## 61  0.937281950     1       1
## 62  0.826394785     1       1
## 63  0.990549125     1       1
## 64  0.831643786     1       1
## 65  0.924628066     1       1
## 66  0.938475497     1       1
## 67  0.709608992     1       1
## 68  0.937880755     1       1
## 69  0.934427707     1       1
## 70  0.956261939     1       1
## 71  0.645281900     1       1
## 72  0.712872076     1       1
## 73  0.946067848     1       1
## 74  0.623698359     1       1
## 75  0.984943871     1       1
## 76  0.861526496     1       1
## 77  0.713649752     1       1
## 78  0.786879736     1       1
## 79  0.950371266     1       1
## 80  0.502777804     1       0
## 81  0.979912101     1       1
## 82  0.876766769     1       1
## 83  0.986226115     1       1
## 84  0.889363445     1       1
## 85  0.669652011     1       1
## 86  0.878246188     1       1
## 87  0.705308717     1       1
## 88  0.839949963     1       1
## 89  0.968547818     1       1
## 90  0.629382032     1       1
## 91  0.891668824     1       1
## 92  0.455085592     1       0
## 93  0.560314411     1       0
## 94  0.815477414     1       1
## 95  0.941601282     1       1
## 96  0.071775006     1       0
## 97  0.708290780     1       1
## 98  0.135421782     1       0
## 99  0.805481190     1       1
## 100 0.653637391     1       1
## 101 0.848779148     1       1
## 102 0.123474673     1       0
## 103 0.936342686     1       1
## 104 0.850755617     1       1
## 105 0.960275266     1       1
## 106 0.818463505     1       1
## 107 0.667207947     1       1
## 108 0.594969886     1       0
## 109 0.967872009     1       1
## 110 0.929599016     1       1
## 111 0.758644391     1       1
## 112 0.813984485     1       1
## 113 0.939645367     1       1
## 114 0.694312821     1       1
## 115 0.889074347     1       1
## 116 0.993687511     1       1
## 117 0.751485538     1       1
## 118 0.754618800     1       1
## 119 0.988667654     1       1
## 120 0.619401811     1       1
## 121 0.211874128     1       0
## 122 0.788220473     1       1
## 123 0.976979052     1       1
## 124 0.989603357     1       1
## 125 0.996607980     1       1
## 126 0.989124713     1       1
## 127 0.745490130     1       1
## 128 0.980603333     1       1
## 129 0.972617049     1       1
## 130 0.609703485     1       1
## 131 0.972075099     1       1
## 132 0.959742735     1       1
## 133 0.918654579     1       1
## 134 0.909037570     1       1
## 135 0.980436813     1       1
## 136 0.952627600     1       1
## 137 0.948219408     1       1
## 138 0.748382589     1       1
## 139 0.320409689     1       0
## 140 0.032023802     1       0
## 141 0.977913666     1       1
## 142 0.725456365     1       1
## 143 0.989478027     1       1
## 144 0.745660544     1       1
## 145 0.932606157     1       1
## 146 0.687176519     1       1
## 147 0.951303317     1       1
## 148 0.993732927     1       1
## 149 0.970261435     1       1
## 150 0.942316651     1       1
## 151 0.355917966     1       0
## 152 0.673763430     1       1
## 153 0.708680217     1       1
## 154 0.903762452     1       1
## 155 0.975453057     1       1
## 156 0.761533292     1       1
## 157 0.974330788     1       1
## 158 0.939641260     1       1
## 159 0.056007279     1       0
## 160 0.705585506     1       1
## 161 0.805795091     1       1
## 162 0.964537461     1       1
## 163 0.952828458     1       1
## 164 0.588731967     1       0
## 165 0.588731967     1       0
## 166 0.003485002     0       0
## 167 0.005708429     0       0
## 168 0.086787249     0       0
## 169 0.074737657     0       0
## 170 0.017197738     0       0
## 171 0.569382403     0       0
## 172 0.540952793     0       0
## 173 0.564244395     0       0
## 174 0.242141594     0       0
## 175 0.006001109     0       0
## 176 0.025246326     0       0
## 177 0.081902908     0       0
## 178 0.947178371     0       1
## 179 0.020340209     0       0
## 180 0.070018033     0       0
## 181 0.039473885     0       0
## 182 0.029297824     0       0
## 183 0.930377477     0       1
## 184 0.356492192     0       0
## 185 0.038640870     0       0
## 186 0.521384267     0       0
## 187 0.089874652     0       0
## 188 0.007810792     0       0
## 189 0.626552595     0       1
## 190 0.537437555     0       0
```

```
## 191 0.358795150     0       0
## 192 0.003152268     0       0
## 193 0.058302258     0       0
## 194 0.005308802     0       0
## 195 0.519223348     0       0
## 196 0.006346126     0       0
## 197 0.495176673     0       0
## 198 0.177800135     0       0
## 199 0.006436658     0       0
## 200 0.515054824     0       0
## 201 0.692363393     0       1
## 202 0.014267145     0       0
## 203 0.043894235     0       0
## 204 0.159381017     0       0
## 205 0.002460244     0       0
## 206 0.254293486     0       0
## 207 0.040017552     0       0
## 208 0.089305064     0       0
## 209 0.107682984     0       0
## 210 0.228988951     0       0
## 211 0.456640546     0       0
## 212 0.017609746     0       0
## 213 0.246231022     0       0
## 214 0.239752868     0       0
## 215 0.077860187     0       0
## 216 0.092107940     0       0
## 217 0.493852386     0       0
## 218 0.007047834     0       0
## 219 0.020145033     0       0
## 220 0.059828902     0       0
## 221 0.020341540     0       0
## 222 0.002417891     0       0
## 223 0.830097987     0       1
## 224 0.002641477     0       0
## 225 0.020246501     0       0
## 226 0.015547078     0       0
## 227 0.057407206     0       0
## 228 0.073121493     0       0
## 229 0.770332875     0       1
## 230 0.275288082     0       0
## 231 0.959228891     0       1
## 232 0.006939516     0       0
## 233 0.027057325     0       0
## 234 0.007302252     0       0
## 235 0.008390043     0       0
## 236 0.275683737     0       0
## 237 0.235591268     0       0
## 238 0.067087799     0       0
## 239 0.097332174     0       0
## 240 0.263376776     0       0
## 241 0.034903522     0       0
## 242 0.576080244     0       0
## 243 0.075641923     0       0
## 244 0.008579446     0       0
## 245 0.035315342     0       0
## 246 0.338255946     0       0
## 247 0.055127657     0       0
## 248 0.096110432     0       0
## 249 0.547909185     0       0
## 250 0.058800597     0       0
## 251 0.001322545     0       0
## 252 0.005849818     0       0
## 253 0.069977170     0       0
## 254 0.036616490     0       0
## 255 0.872442380     0       1
## 256 0.012624733     0       0
## 257 0.004288839     0       0
## 258 0.038604952     0       0
## 259 0.560646242     0       0
## 260 0.602815008     0       1
## 261 0.121915806     0       0
## 262 0.705204571     0       1
## 263 0.004967538     0       0
## 264 0.392801153     0       0
## 265 0.063296586     0       0
## 266 0.186408323     0       0
## 267 0.155576951     0       0
## 268 0.274104132     0       0
## 269 0.006583806     0       0
## 270 0.009475418     0       0
## 271 0.308700864     0       0
## 272 0.504847238     0       0
## 273 0.100340147     0       0
## 274 0.503800659     0       0
## 275 0.051362462     0       0
## 276 0.232649690     0       0
## 277 0.024497236     0       0
## 278 0.678635588     0       1
## 279 0.798374290     0       1
## 280 0.011717651     0       0
## 281 0.214662848     0       0
## 282 0.727559588     0       1
## 283 0.682070914     0       1
## 284 0.693470861     0       1
## 285 0.030229402     0       0
## 286 0.008946191     0       0
## 287 0.884745220     0       1
## 288 0.663155881     0       1
## 289 0.029668353     0       0
## 290 0.185992676     0       0
## 291 0.387641151     0       0
## 292 0.037252329     0       0
## 293 0.099119546     0       0
## 294 0.508658524     0       0
## 295 0.162959962     0       0
## 296 0.005954720     0       0
## 297 0.689997348     0       1
## 298 0.029984503     0       0
## 299 0.318637083     0       0
## 300 0.791228983     0       1
## 301 0.016706652     0       0
## 302 0.025131312     0       0
## 303 0.904212129     0       1
```

```
sum(DF1[2])
```

```
## [1] 165
```

```
sum(DF1[3])
```

```
## [1] 163
```

```
#dodanie predykcji do poczatkowej tabeli
logis <- cbind(logis, C=DF1[3])
write.table(logis, file="lab1result.csv",append=TRUE,sep=";",row.names=TRUE,col.names=TRUE,dec = ",")
```

```
## Warning in write.table(logis, file = "lab1result.csv", append = TRUE, sep =
## ";", : dołączanie nazw kolumn do pliku
```

```
C<-confusionMatrix(factor(DF1$predict), factor(DF1$NAJ.y), positive='1')
C
```

```
## Confusion Matrix and Statistics
## 
##           Reference
## Prediction   0   1
##          0 118  22
##          1  20 143
## 
##                Accuracy : 0.8614
##                  95% CI : (0.8173, 0.8982)
##     No Information Rate : 0.5446
##     P-Value [Acc > NIR] : <2e-16
## 
##                   Kappa : 0.7209
## 
##  Mcnemar's Test P-Value : 0.8774
## 
##             Sensitivity : 0.8667
##             Specificity : 0.8551
##          Pos Pred Value : 0.8773
##          Neg Pred Value : 0.8429
##              Prevalence : 0.5446
##          Detection Rate : 0.4719
##    Detection Prevalence : 0.5380
##       Balanced Accuracy : 0.8609
## 
##        'Positive' Class : 1
## 
```

```
"tabela dobroci klasyfikacji"
```

```
## [1] "tabela dobroci klasyfikacji"
```

```
T<-table(predykacja=DF1$predict,prawdziwe=DF1$NAJ.y)
T
```

```
##           prawdziwe
## predykacja   0   1
##          0 118  22
##          1  20 143
```

```
"obliczanie b??du predykcji"
```

```
## [1] "obliczanie b??du predykcji"
```

```
blad<-1-sum(DF1$predict==DF1$NAJ.y)/nrow(DF1)
"b??d klasyfikacji"
```

```
## [1] "b??d klasyfikacji"
```

```
blad
```

```
## [1] 0.1386139
```

```
WYN<-c(AUC$A,C$overall[[1]],C$overall[[3]],C$overall[[4]],C$overall[[6]],blad,C$byClass[[1]],C$byClass[[2]],C$byClass[[3]],C
$byClass[[4]],LR$`Pr(>Chisq)`[[2]],WALD$`Pr(>F)`[[2]],AIC(NAJ))
WYNNAZ<-c("auc","accuracy","95% CIlower","95% CI upper","p-value","all error","sensitivity","specificity","pos pred valu
e","neg pred value","p for LR","p for WALD","AIC")
D1<-data.frame(WYNNAZ,round(WYN,4))
colnames(D1)<-c("adj. statistic","value")
D1
```

```
##    adj. statistic    value
## 1             auc   0.9232
## 2        accuracy   0.8614
## 3     95% CIlower   0.8173
## 4    95% CI upper   0.8982
## 5         p-value   0.0000
## 6       all error   0.1386
## 7     sensitivity   0.8667
## 8     specificity   0.8551
## 9   pos pred value   0.8773
## 10 neg pred value   0.8429
## 11       p for LR   0.0000
## 12     p for WALD   0.0000
## 13            AIC 234.3206
```

```
#raporty
write.table(D1, file="tab.csv",append=TRUE,sep=";",row.names=FALSE,col.names=TRUE,dec = ",")
```

```
## Warning in write.table(D1, file = "tab.csv", append = TRUE, sep = ";",
## row.names = FALSE, : dołączanie nazw kolumn do pliku
```

```
write.table(W2, file="tab1.csv",append=TRUE,sep=";",row.names=TRUE,col.names=TRUE,dec = ",")
```

```
## Warning in write.table(W2, file = "tab1.csv", append = TRUE, sep = ";", :
## dołączanie nazw kolumn do pliku
```